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# Paper Code : 21322

## F-422

## B.C.A. (Fifth Semester)

## Examination, 2025-26

## (New Course)

## Paper - BCA-502-N

## Computer Network

*Time : Three Hours]* *[Maximum Marks : 70*

**Note** : Attempt any **five** questions. **All** questions carry **equal marks** i.e. 14 marks symbols used have their usual meanings.

1. (a) How many types of computer networking? Explain each with their architecture.

   (b) Describe the connection oriented and connection less services in detail with an example.



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2. (a) How many layers in OSI reference model? Explain the function of each layer.

   (b) How many types of transmission media? Explain it with an example.

3. (a) Describe the message switching and packet circuit switching with an example.

   (b) How many types of error detection method? Describe any one with an example.

4. (a) Describe the Sliding Window Protocol with an example.

   (b) Differentiate between ALOHA and CSMA multiple access protocol.

5. (a) Explain the Static and Dynamic channel allocation in LANs and MANs with an example.

   (b) Differentiate between Source Routing Bridges, Gateways and Routers.



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6. (a) Describe the congestion control and prevention policies in details.

   (b) What are Network Layer design issue? Explain it in details?

7. (a) How many types of IP address? Explain it with an example.

   (b) What is Domain Name System? Explain the all types of Domain Name System with an example.

8. Write short note on any **four** of the following :

   (a) TCP

   (b) Email: Architecture and Services

   (c) TDM and FDM

   (d) Network Security

   (e) CSMA/CD protocals

   (f) File Transfer Protocal (FTP)



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# Paper Code : 21322

## F-422

**B.C.A. (Fifth Semester) Examination, 2023-24**

**(New Course)**

**Paper : BCA-502-N**

**Computer Network**

*Time : Three Hours ]* *[ Maximum Marks : 70*

**Note** : Attempt any **five** questions. **All** questions carry equal i.e. 14 marks. Symbols used have their usual meanings.

1. (a) What is the primary cause of signal loss in satellite communication?

   (b) Explain the advantages & limitation of Twisted pair wire and fiber optical cable.



2. (a) How is TDM different from FDM? Explain with example.

   (b) List the advantages of layering as seen in TCP/IP architecture.

3. (a) Explain the use of hamming code for error detection with the help of example.

   (b) What is ISDN? Explain different types of ISDN.

4. (a) Explain Bridges, Switches, Repeater and Router.

   (b) What is various transmission impairments in data communication?

5. (a) What is IEEE Standard 802.3? Differentiate Ethernet IEEE Standard 802.4 with Token Ring

   (b) Explain various framing method in Data Link Layer.

6. (a) Explain ALOHA protocol for a multi access Channels.

   (b) Explain the functions of TCP Protocols.

7. (a) Explain Routing algorithm.

   (b) Differentiate Packet Switching with Message Switching.

8. Write short notes on any four of the following:

   (a) OSI Model

   (b) X.25

   (c) HTTPs & DNS

   (d) Go-back-N Protocol

   (e) Firewall

   (f) Types of LAN

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# Paper Code : 21322
# F-422

## B.C.A. (Fifth Semester) Examination, 2022-23

**(New Course)**

**Paper-BCA-502-N**

**COMPUTER NETWORK**

***Time : Three Hours ]*** ***[ Maximum Marks : 70***

**Note** : Attempt any **five** questions. All questions carry equal i.e.14 marks. Symbols are used have their usual meanings.

1. (a) Differentiate LAN, MAN and WAN.

   (b) How OSI reference model is different from TCP/IP model? Explain with the help of blok diagram.

2. (a) Explain the types of various transmission media.



   (b) Describe pacpot circuit switching with the help of suitable diagram.

3. (a) What are the ISDN services?

   (b) Differentiate TDM and FDM.

4. (a) Explain the role of data-link layer. How sliding window protocol works?

   (b) Define the following terms :

   (i) ALOHA

   (ii) ATM

5. (a) What are various types of topologies? Explain each with its proper structure. Also compare them on the basis of merits & demerits. https://www.mjpruonline.com

   (b) How "Routing" is useful in networking? Explain the principle of shortest path routing in detail.

6. (a) Write clear difference among router and switch. What is the importance of repeater in any computer network.

(b) What is the role of firewall? Illustrate in detail.

7. (a) Describe various services & protocols in the transport layer.

(b) Summarize all the network security issues and challenges. How to handle with all surch issues.

8. Write briefly about any **four** of the following :

(i) CSMA protocol

(ii) X.25

(iii) Bridges

(iv) Tunneling

(v) E-mail protocols

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# Paper Code : 21322

## F-422

### B.C.A. (Fifth Semester) Examination, 2021

(New Course)

Paper-BCA-502-N

COMPUTER NETWORK

*Time : Three Hours ]* *[ Maximum Marks : 70*

**Note :** Attempt any **five** questions. **All** questions carry equal i.e. 14 marks. Symbols are used have their usual meanings.

1. What is OSI model? List all the layers of OSI model and also write two function of each layer. 14

2. (a) Write the importance of TDM? What are the applications of TDM? Also write its disadvantages (if any). 7



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(b) Explain Go-back-N sliding window protocol. 7

3. (a) Differentiate between packet switching and circuit switching. 7

(b) What is ATM? What are its benefits? Explain. 7

4. (a) Differentiate between twisted pair, coaxial cable, and fiber optics. 7

(b) What is meant by simplex & half duplex communication system? Give example representing each. 7

5. (a) Describe error-detection and correction techniques. 7

(b) What is cyclic redundancy check? 7

6. (a) Describe the following with respect to transport layer: 7

(i) QOS (Quality of Service).

(ii) Connection Establishment.

(iii) Flow control and buffering.



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(b) What are the following standards: 7

IEEE 802.3, IEEE 802.4, AND IEEE 802.5

How do they differ in terms of network topologies and access methods? Explain.

7. (a) Why is multiplexing needed in data communication system? Briefly explain TDM and FDM. 7

(b) What is a Gateway? How do X.25 gate ways link hosts and LANs? Explain. 7

8. Write short notes on any four: 14

(i) Wireless Communication.

(ii) OSI Model.

(iii) CSMA/CD.

(iv) Domain name system.

(v) Bridge.



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# Paper Code : 21322

## F–422

### B.C.A. (Vth Semester) Examination, 2019–20

### (New Course)

### COMPUTER NETWORK

Paper–BCA–502–N

Time : 3 Hours ] [ Maximum Marks : 70

*Note* :– Attempt any *five* questions. All questions carry equal marks i.e. **14** marks. Symbols are used have their usual meanings.

1. (a) Write a short note on guided and unguided media.

   (b) Define simplex, half duplex and full duplex.

2. (a) Differentiate circuit, packet and message switching.

   (b) Compare TCP and UDP Protocols.



3. (a) Define ISDN. What are the various services and layers of ISDN ?

   (b) Define Flow Control. Explain any one flow control algorithm.

4. (a) What is Routing ? Explain distance vector algorithm with an example.

   (b) Explain the following :

   (i) Hamming Code

   (ii) IEEE Standard 802.3

5. (a) Explain various congestion control techniques.

   (b) Differentiate HDLC and SDLC.

6. (a) What is Bridge ? Explain bridges from 802.X to 802.Y.

   (b) State whether true or false "STOP–and–WAIT ARQ is a type of Go-Back-N ARQ Protocol." Justify.

7. (a) Explain E-mail Protocols.

   (b) What is DNS and IP Addressing ? Explain with example.



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8. Write short notes on the following :

(i) OSI Model

(ii) TDM and FDM

(iii) ATM and X.25

(iv) Network Security

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# Paper Code : 21322

## F–422

## B.C.A. (Fifth Semester) Examination, 2018

### (New Course)

Paper-BCA-502-N

### COMPUTER NETWORK

Time : 3 Hours ] [ Maximum Marks : 70

*Note* :– Attempt any *five* questions. All questions carry equal i.e. 14 marks. Symbols are used have their usual meanings.

1. (a) Explain in detail the OSI model with all its seven layers.
   (b) How is it different with TCP/IP model ?
2. (a) Explain about the error detecting and error correcting codes.
   (b) Explain how connections are established using three-way handshake.
3. (a) Explain the various transmission media for data communication.
   (b) Compare circuit, packet and message switching.



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4. Explain in detail any *four* multiple access protocols.
5. (a) Explain the design issues of network layer.
   (b) Define adaptive and non-adaptive algorithm. Explain in detail any *two* routing algorithm.
6. (a) Explain briefly any *two* types of Multiplexing.
   (b) Explain briefly the ATM Service classes.
7. (a) Write the advantage and disadvantages of optical fiber.
   (b) Describe the functions of firewall, gateways and bridges.
8. Write short notes on the following :
   (a) Network topologies
   (b) Ethernet
   (c) X.25 network
   (d) E-mail
   (e) Transport Service Primitives.

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